APPENDIX 5
The Ecological Footprint Explained

Everybody (from a single individual to a whole city or country) has an impact on the Earth, because they consume the products and services of nature. Their ecological impact corresponds to the amount of nature they occupy to keep them going. In this report, we quantify, nation by nation, the biologically productive areas necessary to continuously provide their resource supplies and absorb their wastes, using prevailing technology. In other words, we calculate the "ecological footprints" of these countries.

> Ecological Footprint accounts provide a conservative estimate of humanity's pressure on global ecosystems. They represent the biologically productive area required to produce the food and wood people consume, to supply space for infrastructure, and to absorb the greenhouse gas carbon dioxide (CO2) emitted from burning fossil fuels.
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> The Ecological Footprint is expressed in "global acres." Each global acre corresponds to one acre [hectares] of biologically productive space with world average productivity.
>
> (http://www.rprogress.org/media/releases/021125_efnations.html)

The ecological footprint of an average Canadian, i.e. the amount of land required from nature to support each individual's present consumption, adds up to over 7 hectares or 2.8 acres.

The ecologically productive land available to each person on Earth has decreased over the last century. At the moment there is, on average, 1.8 hectares or 0.7 acres (about one city block) biologically global hectares per person. In contrast, the land appropriated by richer countries has increased. This means that if everyone on Earth lived like the average Canadian, we'd need at least five earths to provide all the material and energy essentials we currently use.

If the world's population continues to grow as anticipated, by the year 2030 there will be 10 billion people, each of whom will have an average of only 0.9 hectares or 0.36 acres of productive land available, assuming there's no further soil degradation. This shows the pressure of population size on nature's productivity.

Increasing density in cities can lead to lower land use requirements, not only because of a reduction in the built environment, but -also because of lifestyles which are less energy-intensive. For example, a recent study of the San Francisco area found that when residential density was doubled, private transportation was reduced by 20 to 30 percent. It's also been shown that residential heating requirements can be reduced significantly if housing is grouped rather than free-standing.

Our challenge is to find a way to balance human consumption and nature's limited productivity in order to ensure that our communities are sustainable locally, regionally and globally. We don't have a choice about whether to do this, but we can choose how we do it. In fact, many people concerned with these issues believe that if we choose wisely now, there's still time for us to make our communities more sustainable, and at the same time improve our quality of life.

There are three key requirements for developing a sustainable community

(a) Ecological health. Use nature's productivity without damaging it.
(b) Community health. Foster social well-being-through the promotion of fairness, equity and

cooperation.
(c) Individual health. Secure food, shelter, health care, education etc. for everyone.

This means working to integrate environmental, economic and social policies so that economic success, ecological integrity and social health become compatible.

In order to make our communities more liveable and sustainable we can work towards change at the personal, urban and commercial levels.

AT HOME WE CAN

- Start composting.
- Use more energy-efficient light bulbs, shower heads etc.
- Switch to forms of recreation and tourism which have a low impact on the environment.
- Grow some of our own food.
- Live closer to work (or the other way around).
- Use bicycles and public transport rather than cars.
- Buy items made or grown locally rather than far away.

Households can start by reducing their resource consumption. At the urban level we must develop an infrastructure that leaves options open, rather than one which dictates resource-intensive lifestyles for our own and future generations. Along with these lifestyle changes, there must be changes in our economies.

CITIES AND TOWNS CAN

- Plan attractive increased population density areas such as town centres and urban villages instead of accommodating further sprawl.
- Offer living, working and shopping spaces in integrated neighbourhoods.
- Reallocate urban space to encourage decreased use of cars. (e.g. reduce road and parking space) and increased use of public transport, bicycles and walking (e.g. build bicycle speedways and attractive pedestrian areas).
- Encourage the planting of trees and green spaces.
- Establish urban land-trusts to give the community more control over land use.
- Promote various kinds of affordable high density housing such: as secondary suites and cooperatives.
- Introduce housing construction guidelines which minimize the consumption of resources, and develop comprehensive waste reduction systems which include municipal resource reuse and reduction schemes.

IN DOING BUSINESS WE CAN

- Rely on using locally available resources rather than imported ones.
- Regain local control over production and distribution of those resources.
- Secure local needs so that the long term livelihood of a region can be protected without compromising the livelihoods of other people in other regions.
- Charge the true costs for private transportation, pollution and resource use.
- Support community-based non-cash, volunteer and mutual aid networks.
- Encourage ecologically sound businesses.
- Offer tax breaks and other incentives for encouraging sustainable lifestyles, and tax and regulate unsustainable: behaviour.

This approach differs from today's global economy which .favours urban industrial centres, and requires the support and involvement of people in each sector of society.

We can all make a difference. Influential groups are:

Politicians (MPs, MLAs, City Councillors, etc) can initiate or support sustainability programs and projects, particularly at the infrastructure level. They can set up screening processes which will take ecological impact into account when assessing a budget or project, and they can encourage the use of the concept of sustainability by the government. They can persuade their parties to develop sustainability strategies, involve the public, and discuss the dilemmas being faced. They can support community groups working towards sustainable societies.

Administrators and planners, who can help politicians write, appropriate legislation and ensure that existing policies are followed. They too can involve the public, present them with the dilemmas and invite input. They can encourage people to participate in shaping the future of their community, and support and assist community groups making positive contributions to society.

The general public, which is all of us - possibly the most important group! We can look at our life styles, think about what's important to us and start family and friends thinking too. Let's get involved and participate in community and municipal groups. Write and talk to politicians at a local, regional or national level, and let them know we want to work with them to develop our communities sustainably.

All of us including politicians and planners are consumers of nature's productivity. We must work together to achieve a more sustainable way of living now in order to ensure that resources continue to be available not only for ourselves, but also for future generations. [14]

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[14] This text was copied from: http://www.iisd.ca/linkages/consume/mwfoot.html and http://www.earthday.net/footprint/quiz3.asp